

Tree Meets Transformer: A Hybrid Architecture for Scalable Power Allocation in Cell-Free Networks

Irched Chafaa , Giacomo Bacci , and Luca Sanguinetti 

Dipartimento di Ingegneria dell'Informazione, University of Pisa, 56122 Pisa, Italy

irched.chafaa@ing.unipi.it, {giacomo.bacci, luca.sanguinetti}@unipi.it

Abstract—Power allocation remains a fundamental challenge in wireless communication networks, particularly under dynamic user loads and large-scale deployments. While Transformer-based models have demonstrated strong performance, their computational cost scales poorly with the number of users. In this work, we propose a novel hybrid Tree-Transformer architecture that achieves scalable per-user power allocation. Our model compresses user features via a binary tree into a global root representation, applies a Transformer encoder solely to this root, and decodes per-user uplink and downlink powers through a shared decoder. This design achieves logarithmic depth and linear total complexity, enabling efficient inference across large and variable user sets without retraining or architectural changes. We evaluate our model on the max-min fairness problem in cell-free massive MIMO systems and demonstrate that it achieves near-optimal performance while significantly reducing inference time compared to full-attention baselines.

Index Terms—Binary tree compression, Transformer, deep learning, scalable inference, power allocation, cell-free.

I. INTRODUCTION

Power allocation is a critical task in wireless networks, particularly in cell-free massive MIMO (mMIMO) systems where users are served by distributed access points (APs) without cell boundaries [1]. The objective is to allocate uplink (UL) and downlink (DL) transmission power efficiently to maximize system throughput and fairness among user equipments (UEs), while adapting to dynamic user loads and spatial configurations.

A. Related Work

Traditional optimization algorithms [2]–[4], such as convex solvers and iterative heuristics, provide reliable solutions but suffer from high computational complexity and poor scalability in large and dynamic networks. Deep learning models [5], [6] have emerged as promising alternatives, enabling fast inference and generalization across varying scenarios. However, neural architectures for power allocation rely on fixed input dimensions, making them inflexible to changes in the

number of UEs and APs. This dependency requires retraining or architectural redesign whenever the wireless network size varies, limiting their applicability in real-world deployments.

Transformer-based architectures [7], [8] have recently shown strong performance due to their ability to capture global dependencies across UEs and APs, while offering flexibility with respect to network size during both training and inference. However, their quadratic complexity with respect to sequence length (number of UEs/APs) [9] remains a bottleneck for large-scale deployments. To address this, efficient Transformer variants with linear attention complexity, such as CosFormer and Performer, have been proposed [10].

In our latest work [11], we introduced a modified CosFormer architecture to jointly predict transmission powers and AP clusters serving each UE. While this model achieves linear complexity with respect to the number of UEs, it still requires full attention computation across all users. Moreover, its reliance on dense inter-user attention makes it less interpretable and harder to deploy in modular or distributed settings.

B. Contributions

To overcome these limitations, we propose a novel hybrid Tree-Transformer architecture for scalable and expressive per-user power allocation. Our model compresses user features into a global root representation using a binary tree structure [12], applies a Transformer encoder [9] solely to this root, and decodes UL and DL power levels for each user via a shared decoder. Through this design, the model fuses local user features with the global state of the wireless network, enabling accurate per-user power predictions that generalize well across varying network sizes and layouts.

We evaluate our model on the max-min fairness problem in cell-free mMIMO systems and compare it against full-attention baselines, including the standard Transformer, CosFormer and Performer-based models. Our results show that the hybrid model achieves near-optimal performance while significantly reducing inference time and computation complexity.

In summary, the main contributions of this paper are as follows:

- We introduce a hybrid Tree-Transformer architecture that combines binary tree compression, root-level attention, and shared decoding for scalable power allocation.
- We demonstrate that the model achieves logarithmic depth and linear total complexity, with Transformer cost independent of user count.

This work was supported by the Smart Networks and Services Joint Undertaking (SNS JU) under the European Union's Horizon Europe research and innovation program under Grant Agreement No 101192369 (6G-MIRAI), by the Italian Ministry of Education and Research (MUR) in the framework of the FoReLab Project (Departments of Excellence), by the HORIZON-JU-SNS-2022 EU project TIMES under grant no. 101096307, and by the European Union under the Italian National Recovery and Resilience Plan (NRRP) of NextGenerationEU, partnership on "Telecommunications of the Future" (PE00000001 – Program "RESTART", Structural Project 6GWINET, Cascade Call SPARKS).

- We show that the model generalizes to varying network settings, and supports modular, interpretable deployment.
- We validate the model on various scenarios, achieving near-optimal performance with significantly reduced computational overhead.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a cell-free mMIMO system (Fig. 1), where K single-antenna UEs are served by L APs with N antennas each. The APs coordinate via a fronthaul network and a central processing unit (CPU) for joint processing and power allocation. The standard time division duplexing (TDD) protocol of cell-free mMIMO is used [3], where the τ_c available channel uses are employed for: (i) UL training phase (τ_p); (ii) DL payload transmission (τ_d); and (iii) UL payload transmission (τ_u). Clearly, $\tau_c \geq \tau_p + \tau_d + \tau_u$. We consider a narrowband channel model and assume that the channel remains constant within a coherence block. We denote the channel vector between the AP l and UE k with $\mathbf{h}_{lk}$, and model it as [3]:

$$\mathbf{h}_{lk} = \sqrt{\beta_{lk}} \mathbf{R}_{lk}^{1/2} \mathbf{g}_{lk}, \quad (1)$$

where β_{lk} is the large-scale fading, accounting for path loss and shadowing, $\mathbf{R}_{lk} \in \mathbb{C}^{N \times N}$ is the spatial correlation matrix at AP l , and $\mathbf{g}_{lk} \sim \mathcal{N}_C(\mathbf{0}, \mathbf{I}_N)$ is an i.i.d. complex Gaussian vector representing the small-scale fading, where $\mathbf{I}_N$ is the $N \times N$ identity matrix. We assume that the channels $\{\mathbf{h}_{lk}; l = 1, \dots, L\}$ are independent and call $\mathbf{h}_k = [\mathbf{h}_{1k}^T, \dots, \mathbf{h}_{Lk}^T]^T \in \mathbb{C}^{LN}$ the collective channel from all APs to UE k .

The CPU computes the estimate of $\mathbf{h}_k$ on the basis of received pilot sequences transmitted during the training phase [3]. The minimum mean square error (MMSE) estimate is $\hat{\mathbf{h}}_k = [\hat{\mathbf{h}}_{1k}^T, \dots, \hat{\mathbf{h}}_{Lk}^T]^T$ with [3]

$$\hat{\mathbf{h}}_{lk} = \mathbf{R}_{lk} \mathbf{Q}_{lk}^{-1} \left(\mathbf{h}_{lk} + \frac{1}{\tau_p \rho} \mathbf{n}_{lk} \right) \sim \mathcal{N}_C(\mathbf{0}_N, \Phi_{lk}), \quad (2)$$

where ρ is the UL pilot power of each UE, $\mathbf{n}_{lk} \sim \mathcal{N}_C(\mathbf{0}_N, \sigma^2 \mathbf{I}_N)$ is the thermal noise, and $\Phi_{lk} = \mathbf{R}_{lk} \mathbf{Q}_{lk}^{-1} \mathbf{R}_{lk}$, where $\mathbf{Q}_{lk} = \mathbf{R}_{lk} + \frac{\sigma^2}{\tau_p \rho} \mathbf{I}_N$. Hence, $\hat{\mathbf{h}}_k \sim \mathcal{N}_C(\mathbf{0}_{LN}, \Phi_k)$, with $\Phi_k = \text{diag}(\Phi_{1k}, \dots, \Phi_{Lk})$. Note that the method proposed in this paper can be applied to other channel estimation schemes.

A. Uplink and Downlink Transmissions

To detect the data of UE k in the UL, the CPU selects an arbitrary receive combining vector $\mathbf{v}_k \in \mathbb{C}^{LN}$ based on all the collective channel estimates $\{\hat{\mathbf{h}}_k; k = 1, \dots, K\}$. An achievable spectral efficiency (SE) of UE k is given by [3]:

$$\text{SE}_k^{\text{UL}} = \frac{\tau_u}{\tau_c} \log_2(1 + \text{SINR}_k^{\text{UL}}), \quad (3)$$

with the effective signal-to-interference-plus-noise ratio (SINR) defined as

$$\frac{p_k^{\text{UL}} |\mathbb{E}\{\mathbf{v}_k^H \mathbf{h}_k\}|^2}{\sum_{i=1}^K p_i^{\text{UL}} \mathbb{E}\{|\mathbf{v}_k^H \mathbf{h}_i|^2\} - p_k^{\text{UL}} |\mathbb{E}\{\mathbf{v}_k^H \mathbf{h}_k\}|^2 + \sigma^2 \mathbb{E}\{\|\mathbf{v}_k\|^2\}}, \quad (4)$$

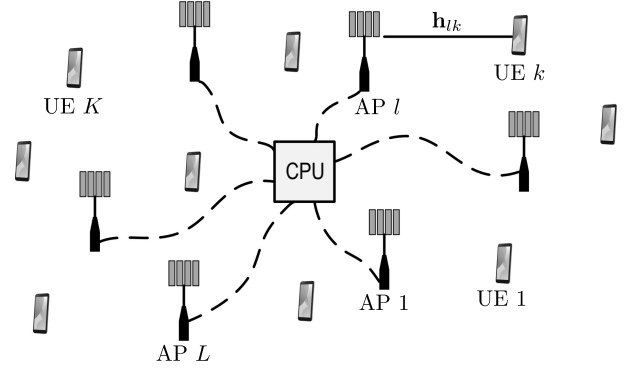


Fig. 1: Illustration of a cell-free mMIMO network. Users are served by APs without cell boundaries.

where p_k^{UL} is UE k 's UL transmit power. The expectation $\mathbb{E}\{\cdot\}$ is taken with respect to all sources of randomness. Although the bound in (3) is valid for any combining vector, we consider the MMSE combiner, given by [3]:

$$\mathbf{v}_k = \left(\sum_{k=1}^K p_k^{\text{UL}} \hat{\mathbf{h}}_k \hat{\mathbf{h}}_k^H + \mathbf{Z} \right)^{-1} \hat{\mathbf{h}}_k, \quad (5)$$

where $\mathbf{Z} = \sum_{k=1}^K p_k^{\text{UL}} [\text{diag}(\mathbf{R}_{1k}, \dots, \mathbf{R}_{Lk}) - \Phi_k] + \sigma^2 \mathbf{I}_{LN}$.

In the DL, the CPU coordinates the APs to transmit signals to the UEs. Similarly to UL, an achievable SE of user k is obtained as:

$$\text{SE}_k^{\text{DL}} = \frac{\tau_d}{\tau_c} \log_2(1 + \text{SINR}_k^{\text{DL}}), \quad (6)$$

with the effective SINR defined as

$$\frac{p_k^{\text{DL}} |\mathbb{E}\{\mathbf{h}_k^H \mathbf{w}_k\}|^2}{\sum_{i=1}^K p_i^{\text{DL}} \mathbb{E}\{|\mathbf{h}_k^H \mathbf{w}_i|^2\} - p_k^{\text{DL}} |\mathbb{E}\{\mathbf{h}_k^H \mathbf{w}_k\}|^2 + \sigma^2}, \quad (7)$$

where p_k^{DL} is the DL power used by the CPU to serve UE k such that $p_k^{\text{DL}} = \sum_{l=1}^L p_{k,l}^{\text{DL}}$, with $p_{k,l}^{\text{DL}} \in [0, \bar{P}_l^{\text{DL}}]$ being the AP l 's transmit power allocated for user k ; $\bar{P}_l^{\text{DL}}$ is the maximum power per AP; and $\mathbf{w}_k \in \mathbb{C}^{LN}$ is the associated unit-norm precoding vector. The MMSE precoder is used [3], which is given by $\mathbf{w}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$.

B. Problem Formulation

Our goal is to allocate transmission powers $\{p_k^{\text{UL}}, p_k^{\text{DL}}\}$ to ensure SE fairness among users while satisfying system constraints. Specifically, we consider the max-min fairness criterion, which aims to maximize the minimum achievable SE across all UEs. This leads to the following optimization problem [3]:

$$\begin{aligned} & \max_{\{p_k^{\text{UL}} \geq 0\}} \min_k \text{SE}_k^{\text{UL}} \\ & \text{subject to } p_k^{\text{UL}} \leq \bar{P}_k^{\text{UL}} \quad \forall k \end{aligned} \quad (8)$$

where $\bar{P}_k^{\text{UL}}$ is the maximum UL power for user k . Similarly, in the DL we have that:

$$\begin{aligned} & \max_{\{p_k^{\text{DL}} \geq 0\}} \min_k \text{SE}_k^{\text{DL}} \\ & \text{subject to} \quad \sum_{k=1}^K p_k^{\text{DL}} \leq \sum_{l=1}^L \bar{P}_l^{\text{DL}} \end{aligned} \quad (9)$$

where the constraint ensures that the total power allocated to all UEs does not exceed the total power budget across all APs.

Both optimization problems can be solved using closed-form approximations [4], iterative solvers [2], [3], or deep learning approaches [5]–[7], [13]. However, as discussed earlier, these methods either suffer from high computational cost, require retraining for different network sizes, or scale poorly with the number of users. To address these limitations, we propose a hybrid Tree-Transformer architecture that learns to predict UL and DL power directly from UE and AP positions.

III. PROPOSED HYBRID TREE-TRANSFORMER MODEL

This section details the proposed learning approach for power prediction including: the dataset construction, input processing, model design, and training procedure.

A. Dataset Construction

To train and evaluate the proposed model, we construct a synthetic dataset using typical simulation parameters for cell-free mMIMO systems [3].

a) *Data generation.* For each configuration defined by a pair (K, L) , we generate:

- random bidimensional (2D) positions in a given area for K single-antenna UEs, denoted as $\{\mathbf{u}_k\}_{k=1}^K$, with $\mathbf{u}_k \in \mathbb{R}^2$, and L APs, each equipped with N antennas, denoted as $\{\mathbf{a}_l\}_{l=1}^L$, with $\mathbf{a}_l \in \mathbb{R}^2$;
- channel realizations $\mathbf{h}_{lk}$ based on the model in (1), using spatial correlation matrices and large-scale fading coefficients β_{lk} ;
- optimal UL and DL power allocations computed using the closed-form max-min fairness solution from [4].

b) *Noise Injection.* To simulate realistic deployment conditions and improve generalization, we inject Gaussian noise into the UE and AP positions:

$$\tilde{\mathbf{u}}_k = \mathbf{u}_k + \delta_k, \quad \delta_k \sim \mathcal{N}(0, \sigma_k^2), \quad (10)$$

$$\tilde{\mathbf{a}}_l = \mathbf{a}_l + \delta_l, \quad \delta_l \sim \mathcal{N}(0, \sigma_l^2), \quad (11)$$

where $\sigma_k = \sigma_l = 5$ m controls the noise level [14].

c) *Normalization.* All features (positions and powers) are normalized to the range $[0, 1]$ using min-max scaling [6]. For a feature vector ξ , the normalized value is computed as

$$\xi_{\text{norm}} = \frac{\xi - \min(\xi)}{\max(\xi) - \min(\xi) + \varepsilon}, \quad (12)$$

where ε is a small constant added to avoid division by zero. Each 2D position with spatial coordinates (x, y) is normalized by applying min-max scaling separately to the x - and y -coordinates across all APs or UEs. This preserves geometric relationships such as distances and

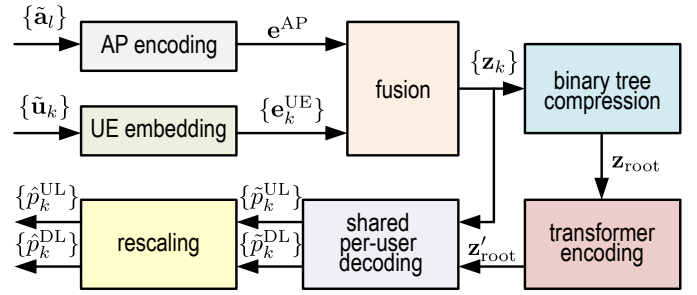


Fig. 2: Block diagram of the proposed model architecture.

angles between nodes while standardizing the input range for stable training. UL and DL powers are normalized globally across all samples to ensure consistent scaling.

d) *Dataset overview.* A total number of 8000 samples for each value of $K \in \{2, 4, 6, 8, 10\}$ and $L = 16$ are used for training. This diversity enables the model to learn robust mapping across varying network sizes, spatial layouts and channel realizations. Each sample includes UE and AP positions as input, and optimal power allocations as output labels. Additionally, 200 samples for each value of $K = 2, 3, \dots, 40$ and $L \in \{1, 4, 9, 16, 25\}$ are used for testing to assess the model's ability to generalize to new configurations.

B. Model Architecture

The proposed hybrid Tree-Transformer model is designed to predict UL and DL power allocations for each UE based on the spatial configuration of UEs and APs. The architecture consists of seven main components: an AP encoder, a UE embedding module, a fusion method, a binary tree compressor, a Transformer encoder applied to the root node, a shared per-user decoder, and a final rescaler. The overall structure is illustrated in Fig. 2.

a) *AP encoder.* The (noisy) coordinates of all APs $\{\tilde{\mathbf{a}}_l\}_{l=1}^L$ are passed through a two-layer multi-layer perceptron (MLP). The first linear layer maps the 2D inputs to a hidden dimension d_{enc} , followed by a rectified linear unit (ReLU) activation [15]. A second linear layer refines the representation while keeping the same size. Hence, the embedding of the l -th AP becomes $\mathbf{e}_l^{\text{AP}} = \text{MLP}(\tilde{\mathbf{a}}_l) \in \mathbb{R}^{d_{\text{enc}}}$, whereas the global AP context is obtained by averaging across them, as $\mathbf{e}^{\text{AP}} = \frac{1}{L} \sum_{l=1}^L \mathbf{e}_l^{\text{AP}}$. This global vector $\mathbf{e}^{\text{AP}} \in \mathbb{R}^{d_{\text{enc}}}$ summarizes the whole distribution of APs in the environment.

b) *UE embedding.* Similarly to the APs, each UE is embedded via the linear transformation $\mathbf{e}_k^{\text{UE}} = \mathbf{W}_{\text{UE}} \tilde{\mathbf{u}}_k + \mathbf{b}_{\text{UE}}$, where $\mathbf{W}_{\text{UE}} \in \mathbb{R}^{d_{\text{enc}} \times 2}$ and $\mathbf{b}_{\text{UE}} \in \mathbb{R}^{d_{\text{enc}}}$ are trainable parameters.

c) *Fusion.* The AP embedding $\mathbf{e}^{\text{AP}}$ is concatenated to each user embedding $\mathbf{e}_k^{\text{UE}}$ to incorporate global context: $\mathbf{z}_k = [(\mathbf{e}_k^{\text{UE}})^T (\mathbf{e}^{\text{AP}})^T]^T \in \mathbb{R}^{2d_{\text{enc}}}$. The set of fused descriptors $\{\mathbf{z}_k\}_{k=1}^K$ serves as the input to both the binary tree compressor and the shared decoder.

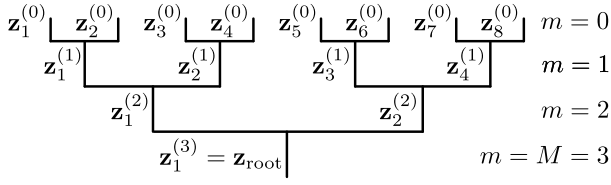


Fig. 3: Example of binary tree compression ($K = 8$).

- d) *Binary tree compressor.* The fused user descriptors $\{\mathbf{z}_k\}_{k=1}^K$ are first projected into a common embedding space of dimension d_{mod} using a linear transformation $\mathbf{z}_k^{(0)} = \mathbf{W}_0 \mathbf{z}_k + \mathbf{b}_0$, where $\mathbf{W}_0 \in \mathbb{R}^{d_{\text{mod}} \times 2d_{\text{enc}}}$ and $\mathbf{b}_0 \in \mathbb{R}^{d_{\text{mod}}}$. They are then hierarchically aggregated using a binary tree structure with $M = \lceil \log_2 K \rceil$ compression stages, where $\lceil \cdot \rceil$ is the ceiling function, as illustrated in Fig. 3 for $K = 8$ ($M = 3$). At each stage $m = 1, \dots, M$, the parent vectors $\mathbf{z}_q^{(m)}$, $q = 1, \dots, 2^{M-m}$, are obtained using the linear transformation $\mathbf{z}_q^{(m)} = \mathbf{W}[(\mathbf{z}_{2q-1}^{(m-1)})^\top (\mathbf{z}_{2q}^{(m-1)})^\top]^\top + \mathbf{b}$, where $\mathbf{b} \in \mathbb{R}^{d_{\text{mod}}}$ is a trainable bias, and $\mathbf{W}$ is a learnable weight matrix, with size $d_{\text{mod}} \times 2d_{\text{mod}}$. If the number of descriptors at any level is odd, a zero-padding vector is appended to enable pairwise merging. The process continues until a single root descriptor $\mathbf{z}_{\text{root}} = \mathbf{z}_1^{(M)}$ is obtained. Since the depth of the tree is logarithmic in K , and the total number of merge operations scales linearly with K , the compression is efficient and scalable. This hierarchical aggregation strategy is inspired by prior work in tree-based encoders [12], [16], enabling global context extraction without quadratic complexity.
- e) *Transformer encoder.* The root descriptor $\mathbf{z}_{\text{root}}$ is further refined using a Transformer encoder [9] with S stacked layers, each comprising a multi-head self-attention block followed by a feed-forward network. Each attention block uses A attention heads, and all internal representations maintain the same dimensionality d_{mod} . Since the encoder operates on a single root descriptor rather than a sequence of tokens, the self-attention mechanism degenerates to multi-head linear projections of this vector. In practice, each head applies a distinct learned transformation, and their concatenation enriches the global representation. This design allows the model to benefit from the expressive diversity of multi-head attention while avoiding the quadratic cost of conventional Transformers. The resulting enriched global context vector $\mathbf{z}'_{\text{root}} \in \mathbb{R}^{d_{\text{mod}}}$ is then used by the decoder at the next stage.
- f) *Shared per-user decoder.* The enriched global context vector $\mathbf{z}'_{\text{root}}$ is broadcast to all UEs and concatenated with their original fused descriptors $\{\mathbf{z}_k\}_{k=1}^K$. This results in a context-aware vector for each user, $\mathbf{z}_k^{\text{dec}} = [(\mathbf{z}_k)^\top (\mathbf{z}'_{\text{root}})^\top]^\top \in \mathbb{R}^{2d_{\text{enc}} + d_{\text{mod}}}$. Each $\mathbf{z}_k^{\text{dec}}$ is then passed through a shared decoder network, implemented as a two-layer MLP with ReLU activation, followed by a sigmoid output [15]. The decoder maps the vector $\mathbf{z}_k^{\text{dec}}$ to a pair

model parameters		training parameters	
AP/UE embedding dimension d_{enc}	32	learning rate	10^{-3}
model dimension d_{mod}	64	batch size	128
number of attention heads A	4	number of epochs	100
number of Transformer layers S	2	optimizer	AdamW [18]

Table I: Model parameters and training setup configuration.

of normalized power values $\tilde{\mathbf{p}}_k = [\tilde{p}_k^{\text{UL}}, \tilde{p}_k^{\text{DL}}] \in [0, 1]^2$, where $\tilde{p}_k^{\text{UL}}$ and $\tilde{p}_k^{\text{DL}}$ denote the normalized predicted UE k 's UL and DL powers. This shared decoding strategy enables parameter efficiency and consistent prediction behavior across users, regardless of the network size (K, L) . By conditioning each prediction on both local user features and the globally refined context, the model captures both fine-grained and holistic patterns in the wireless network for power prediction.

- g) *Rescaler.* A final step involves rescaling the predictions back to their original ranges, by applying de-normalization and rescaling:

$$\hat{p}_k^{\text{UL}} = \Delta_{\text{UL}} \tilde{p}_k^{\text{UL}} + \underline{P}_{\text{UL}}, \quad (13)$$

$$\hat{p}_k^{\text{DL}} = \frac{\tilde{p}_k^{\text{DL}} \sum_{l=1}^L \bar{P}_l^{\text{DL}}}{\sum_{k=1}^K \tilde{p}_k^{\text{DL}}}, \text{ with } \tilde{p}_k^{\text{DL}} = \Delta_{\text{DL}} \tilde{p}_k^{\text{DL}} + \underline{P}_{\text{DL}}, \quad (14)$$

where $\Delta_{\text{UL}} = \bar{P}_{\text{UL}} - \underline{P}_{\text{UL}}$ (resp., $\Delta_{\text{DL}} = \bar{P}_{\text{DL}} - \underline{P}_{\text{DL}}$) is the UL (resp., DL) power range, with $\bar{P}_{\text{UL}}$ (resp. $\underline{P}_{\text{UL}}$) denoting UE k 's maximum (resp., minimum) UL power, and $\bar{P}_{\text{DL}}$ (resp. $\underline{P}_{\text{DL}}$) being the counterpart on the DL.

C. Model Parameters

The hybrid Tree-Transformer model is implemented using PyTorch [17], and configured with the parameters in Table I, selected empirically to balance model expressiveness and computational efficiency. The architecture remains lightweight, enabling scalability across varying user counts while retaining sufficient capacity to learn complex mappings from spatial features to power allocation.

D. Training Setup

The hybrid Tree-Transformer model is trained to minimize the mean square error (MSE) $\mathcal{L}_{\text{MSE}}$ between the normalized predicted powers and the optimal ones obtained from the closed-form solution [4]. By doing so, the model implicitly learns the effects of the channel propagation environment, since the optimal powers embed channel information. As a result, the network is trained to approximate the mapping from the spatial positions of UEs and APs to power allocations that maximize the minimum SE.

The training is performed across multiple values of the number of users K , ensuring that the model is exposed to heterogeneous scenarios and can generalize across different network sizes. Validation is also conducted across varying sizes (K, L) . The specific hyperparameters used during training are summarized in Table I. The best-performing model is saved based on the lowest average validation loss across different network configurations.

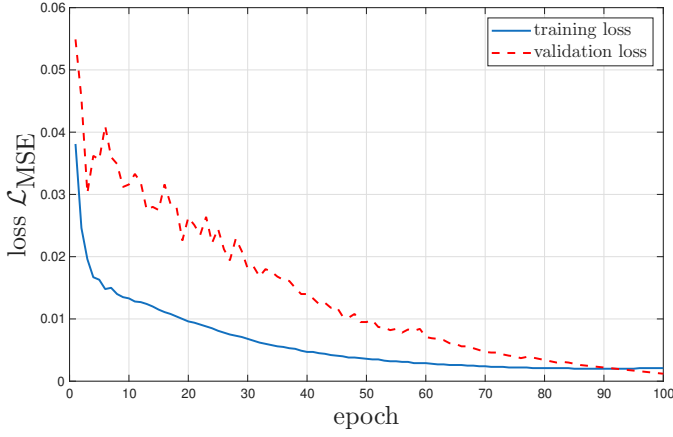


Fig. 4: Evolution of training and validation loss. The model converges rapidly and generalizes well across network configurations.

IV. NUMERICAL RESULTS

In this section, we present numerical results to illustrate the performance of the proposed approach for a cell-free mMIMO system, as described in Sect. II. We consider a network with a coverage area of $500\text{ m} \times 500\text{ m}$, with $N = 4$ antennas per AP. The APs are uniformly deployed within the squared coverage area. The maximum UL transmit power for each user is $\bar{P}_{\text{UL}} = 100\text{ mW}$, whereas the maximum DL transmit power for each AP is $\bar{P}_l^{\text{DL}} = 200\text{ mW}$. We assume $\tau_c = 200$ and set $\tau_p = 10$, $\tau_u = 90$ and $\tau_d = 100$. Large-scale fading coefficients are computed following the 3GPP path-loss model adopted in [4, Sect. III-D] for a 2-GHz carrier frequency, a pathloss exponent of 3.67, a UE-AP height difference of 10 m and a shadow fading $F_{kl} \sim \mathcal{N}_C(0, \alpha^2)$, with $\alpha^2 = 4\text{ dB}$. The shadow fading terms are spatially correlated as in [4, Sect. III-D] to account for the fact that closely located UEs experience similar shadow fading effects. The noise power is $\sigma^2 = -94\text{ dBm}$ [4] with a noise figure $\eta = 7\text{ dB}$ and a bandwidth $B = 20\text{ MHz}$. The dataset and codes for model training are available in [19].

Training performance. Fig. 4 shows the evolution of training and validation loss $\mathcal{L}_{\text{MSE}}$ over epochs. The consistent downward trend in both curves indicates that the model effectively learns to generalize across varying user distributions. The eventual close alignment between training and validation loss suggests minimal overfitting and strong predictive performance. Minor fluctuations in the validation curve are attributed to variability in spatial layouts and channel conditions across samples. Nevertheless, the overall convergence behavior confirms the robustness of the learning process and the model's ability to accurately predict power.

Prediction accuracy. Fig. 5 shows the cumulative distribution functions (CDFs) of the optimal and predicted UL and DL power on the test set, which includes diverse network configurations (K, L) not encountered during training. We can see that the curves of predicted and optimal powers almost completely overlap, demonstrating that the model can generalize effectively to new settings and reproduce the optimal

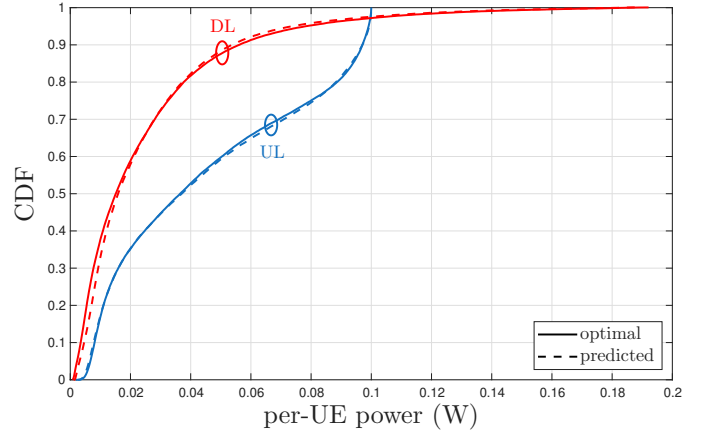


Fig. 5: CDFs of optimal and predicted power on the test set. The predicted curves closely follow the optimal ones, confirming the model's robustness and generalization capability.

UL and DL power distributions with high fidelity, even under noisy input conditions.

Flexibility with network size. To assess the generalization capability of the proposed model, in Fig. 6 we evaluate its SE performance on new network settings with varying numbers of UEs K and APs L , without retraining or modifying the architecture. We can see that the predicted SE curves closely follow the optimal ones with only a small gap: on average, the prediction remains within 0.12 b/s/Hz of the optimal SE in both simulations, confirming that the model can adapt to different network sizes and configurations while maintaining high accuracy. Since the optimal closed-form solution is available and our model achieves comparable SE values, we do not extend the SE comparison to other methods in the literature.

Comparison with Transformer variants. To highlight the efficiency of our proposed model, we compare it against Transformer [9], CosFormer, and Performer [10]. The results are reported in Table II for $K = 40$, $L = 16$ (for the Performer case, we consider $F = 256$ random features). For all variants, the input consists of UE/AP positions, and the output is the predicted UL and DL power, following the same setup as in [8].

Our model restricts attention computation to the compressed binary tree representation, yielding logarithmic depth and linear complexity. Unlike other models, its cost is independent of the wireless network size (K, L) . As a result, it achieves competitive SE while offering the lowest latency among all compared models, as highlighted in bold in Table II. This confirms its suitability for large-scale cell-free mMIMO systems and distributed deployments where local prediction is required. As a reference, the optimal closed-form solution [4] has complexity $\mathcal{O}(K^3)$, which is significantly higher than all learning-based approaches.

V. CONCLUSION

In this work, we propose a Tree-Transformer architecture for power allocation in cell-free mMIMO systems. By com-

metric	our model	Transformer	CosFormer	Performer
parameters	584,354	845,826	102,338	102,338
latency (ms)	3.57	28.46	7.11	25.45
avg. SE (b/s/Hz)	2.01	1.66	1.80	1.81
scalability with K	$\mathcal{O}(K \cdot d_{\text{mod}}^2)$	$\mathcal{O}(K^2 \cdot S \cdot d_{\text{mod}}^2)$	$\mathcal{O}(K \cdot S \cdot d_{\text{mod}}^2)$	$\mathcal{O}(K \cdot S \cdot d_{\text{mod}} \cdot F)$
scalability with L	$\mathcal{O}(L \cdot d_{\text{enc}})$	$\mathcal{O}(L \cdot K \cdot d_{\text{mod}})$	$\mathcal{O}(L \cdot K \cdot d_{\text{mod}})$	$\mathcal{O}(L \cdot K \cdot d_{\text{mod}})$

Table II: Performance comparison of power prediction models using Transformer variants.

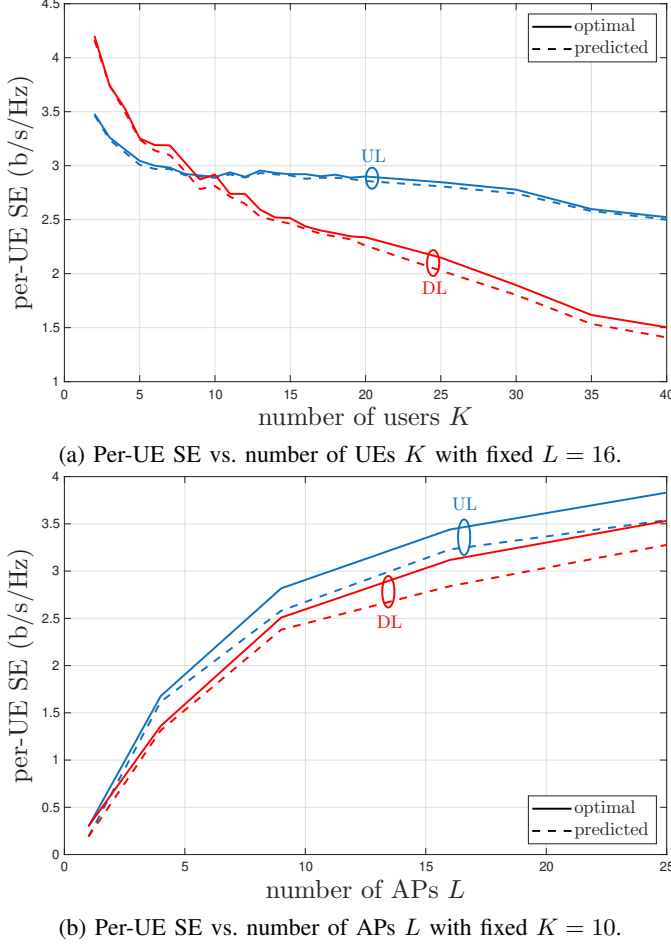


Fig. 6: Comparison of SE trends: (a) varying UE count K , (b) varying AP count L . Predicted SE closely tracks the optimal SE across both scenarios.

binning binary tree compression with Transformer-based global reasoning, the model achieves high prediction accuracy while significantly reducing complexity compared to both standard Transformer variants and the optimal closed-form solution. Numerical results demonstrate that the proposed approach generalizes well to unseen network configurations, remains robust under noisy inputs, and achieves SE values close to the optimal ones with minimal latency. Beyond the immediate performance gains, the Tree-Transformer offers scalability with respect to both the number of UEs and APs, making it suitable for large-scale deployments and distributed implementations.

As a perspective, a promising extension of this work is the integration of distributed learning strategies, where multiple APs collaboratively train or update local models without

centralized coordination. This approach would further enhance scalability and adaptability in large-scale deployments.

REFERENCES

- [1] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, and T. L. Marzetta, "Cell-free massive MIMO versus small cells," *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1834–1850, 2017.
- [2] M. Farooq, H. Q. Ngo, and L.-N. Tran, "Accelerated projected gradient method for the optimization of cell-free massive MIMO downlink," in *Proc. Intl. Symp. Personal, Indoor and Mobile Radio Commun. (PIMRC)*, London, UK, 2020.
- [3] Ö. T. Demir, E. Björnson, and L. Sanguinetti, "Foundations of user-centric cell-free massive MIMO," *Foundations and Trends® in Signal Processing*, vol. 14, no. 3-4, pp. 162–472, 2021.
- [4] L. Miretti, R. L. G. Cavalcante, S. Stańczak, M. Schubert, R. Böhnke, and W. Xu, "Closed-form max-min power control for some cellular and cell-free massive MIMO networks," in *Proc. IEEE Veh. Technol. Conf.*, Helsinki, Finland, 2022.
- [5] D. Kim, H. Jung, and I.-H. Lee, "A survey on deep learning-based resource allocation schemes," in *Proc. Intl. Conf. Information and Commun. Technol. Convergence (ICTC)*, Jeju Island, South Korea, 2023.
- [6] Q. Mao, F. Hu, and Q. Hao, "Deep learning for intelligent wireless networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2595–2621, 2018.
- [7] A. K. Kocharlakota, S. A. Vorobyov, and R. W. Heath Jr, "Pilot contamination aware transformer for downlink power control in cell-free massive MIMO networks," *arXiv preprint arXiv:2411.19020*, 2024.
- [8] I. Chafaa, G. Bacci, and L. Sanguinetti, "Transformer-based power optimization for max-min fairness in cell-free massive MIMO," *IEEE Wireless Commun. Lett.*, vol. 14, no. 8, pp. 2316–2320, 2025.
- [9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Conf. Neural Inf. Process. Systems (NIPS)*, Long Beach, CA, USA, 2017.
- [10] Y. Elouargui, M. Zyate, A. Sassioui, M. Chergui, M. El Kamili, and M. Ouzif, "A comprehensive survey on efficient transformers," in *Proc. Intl. Conf. Wireless Netw. and Mobile Commun. (WINCOM)*, Istanbul, Turkey, 2023.
- [11] I. Chafaa, G. Bacci, and L. Sanguinetti, "Linear attention for joint power optimization and user-centric clustering in cell-free networks," *IEEE Trans. Wireless Commun.*, 2025, submitted.
- [12] J. Harer, C. Reale, and P. Chin, "Tree-transformer: A transformer-based method for correction of tree-structured data," *arXiv preprint arXiv:1908.00449*, 2019.
- [13] Y.-J. Choi, J.-H. Yu, H.-Y. Jeong, J.-E. Kim, and H.-K. Song, "Deep learning-based power allocation for cell-free massive MIMO networks with adaptive access point power control," in *Proc. Intl. Conf. Ubiquitous and Future Networks (ICUFN)*, Lisbon, Portugal, 2025, pp. 744–749.
- [14] C. Xue, P. Psimoulis, Q. Zhang, and X. Meng, "Analysis of the performance of closely spaced low-cost multi-GNSS receivers," *Applied Geomatics*, vol. 13, no. 3, pp. 415–435, 2021.
- [15] A. D. Rasamoelina, F. Adjailia, and P. Sinčák, "A review of activation function for artificial neural network," in *Proc. IEEE World Symp. Applied Machine Intell. & Informatics (SAMI)*, Herlany, Slovakia, 2020.
- [16] K. S. Tai, R. Socher, and C. D. Manning, "Improved semantic representations from tree-structured long short-term memory networks," *arXiv preprint arXiv:1503.00075*, 2015.
- [17] A. P. et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Intl. Conf. Neural Information Processing Systems*, Vancouver, Canada, 2019.
- [18] P. Zhou, X. Xie, Z. Lin, and S. Yan, "Towards understanding convergence and generalization of AdamW," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 6486–6493, 2024.
- [19] I. Chafaa, "Hybrid Tree-Transformer model," <https://github.com/irchchaf/Hybrid-Tree-Transformer-Model>, 2025, GitHub repository.